<u>Hand-in 2:</u>

<u>Product Development and Engineering Design 151-3202-00L</u>

Group 7

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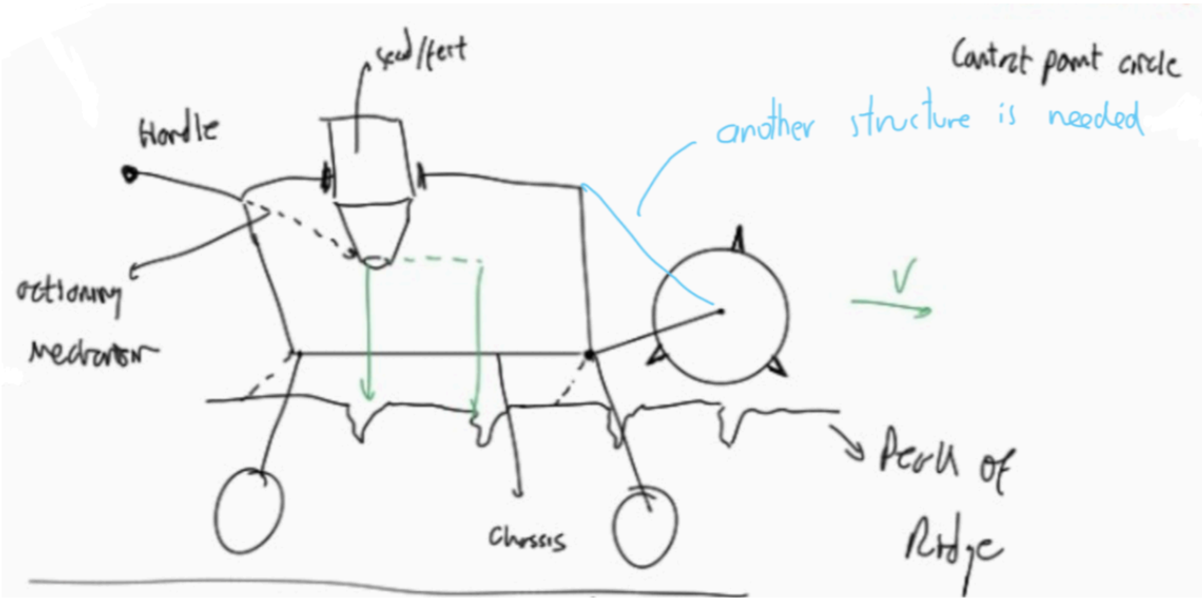
1. Concept Generation - Brainwriting & Design Heuristics Cards

Three members of the team have applied the Brainwriting in a 3-2-5 format developing 6 ideas in total. We can see in the different designs annotations and drawings with different colors representing the cooperative work of the three members. During the design the heuristic cards have been used.

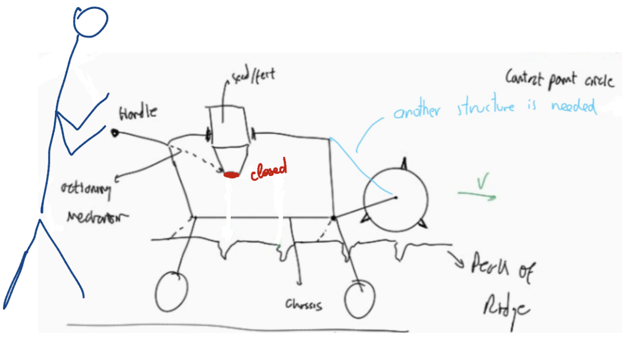
1.1. Design A by Alex

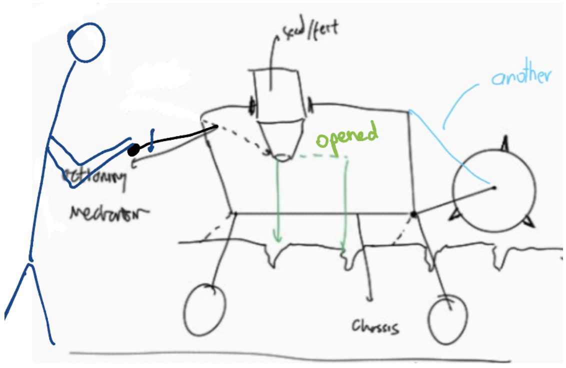
This concept is a four-wheeled cart equipped with a front roller that penetrates the soil to create planting holes. A manually actuated trap, positioned by the user directly above each hole, releases the seeds and fertilizer. After deposition, an angled closing element pushes the soil back over the hole, ensuring proper coverage without additional user effort.



Heuristic Cards
- Card 2: add motion = we added the rotating wheel so it can continue unlimitedly
- Card 29: create system = multi-stage process to complete the function
- Card 38: impose hierarchy on functions = first digging the hole, then planting the seed and finally covering the hole
- Card 71: use human-generated power = using the kinetic of the user to push the machine without the help of a motor

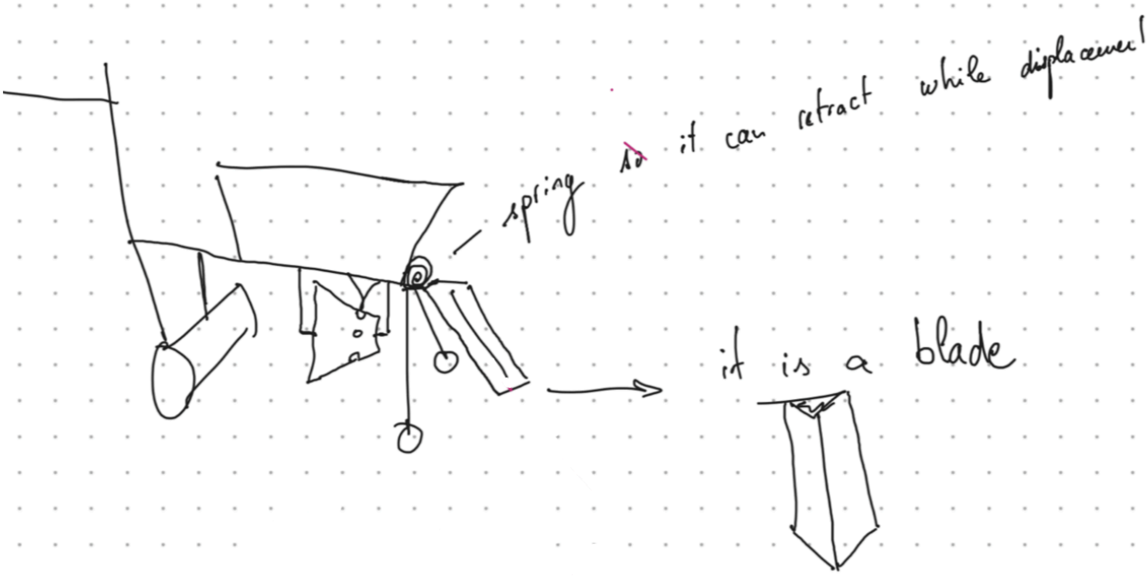
In the last update we used card 40 to incorporate user input :





1.2 Design B by Antoine

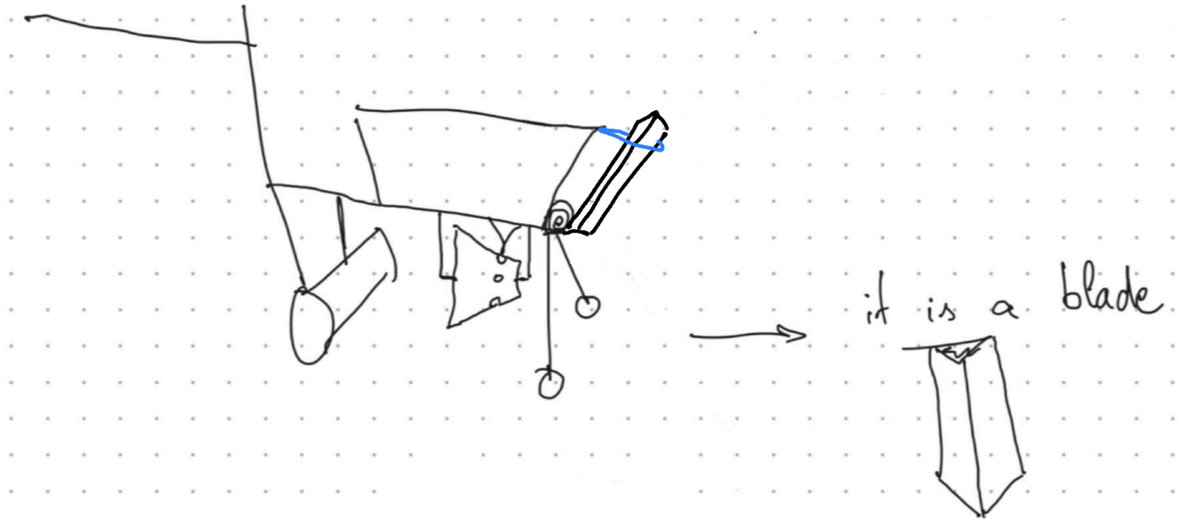
This concept is a wheelbarrow-like device that uses a front blade to open the soil and create a planting furrow, followed by a rear blade that closes the soil after deposition. A mechanical transmission links the wheels to a rotating cone, which meters and releases seeds and fertilizer. As the device moves forward, the cone rotates and drops the inputs directly into the opened soil before it is covered again.



Heuristic Cards

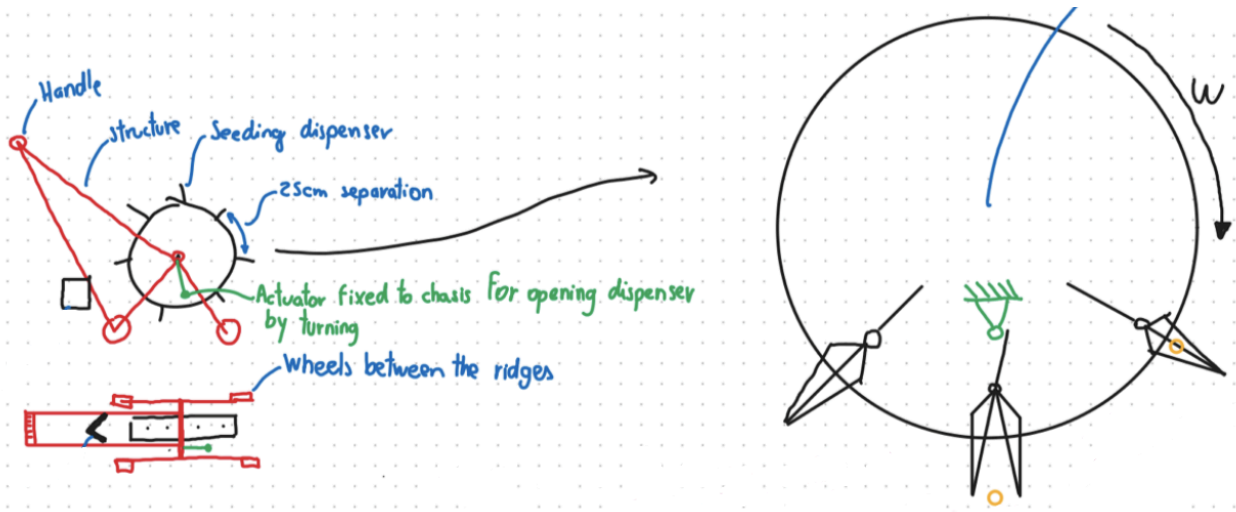
- Card 5: adjust function through movement = move the cone a bit further to choose another mode and now plant 3 seeds every 75cm.
- Card 15: attach product to user = the machine can be attached to a bike, so the user can move more freely
- Card 29: create system = multi-stage process to complete the function
- Card 33: expose interior = inner parts are visible enabling the visual control of user
- Card 38: impose hierarchy on functions = first digging the hole, then planting the seed and finally covering the hole
- Card 71: use human-generated power = using the kinetic of the user to push the machine without the help of a motor

In the last update we used card 36 to incorporate folding capacity :



1.3 Design C by Luis

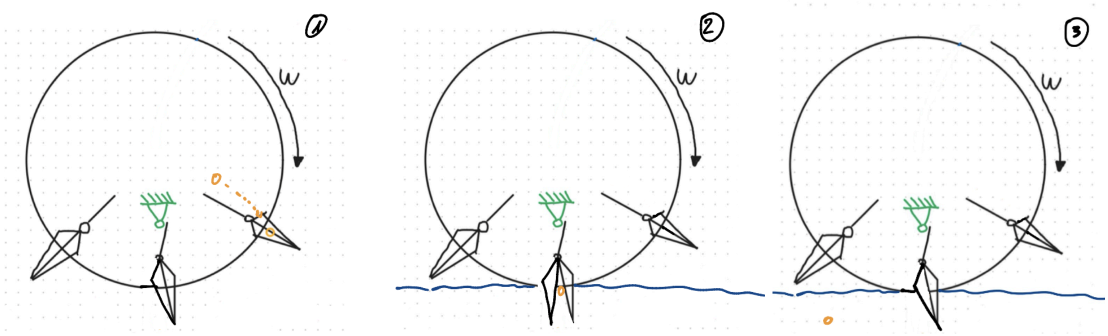
This concept is a four-wheeled cart equipped with a central rotating wheel fitted with spikes. As the device moves forward, the spikes penetrate the soil to create planting holes. During this motion, the mechanism opens to release seeds or fertilizer directly into the hole. As the wheel continues rotating, the spikes exit the soil and the cycle repeats, enabling continuous planting at regular intervals.



Heuristic Cards

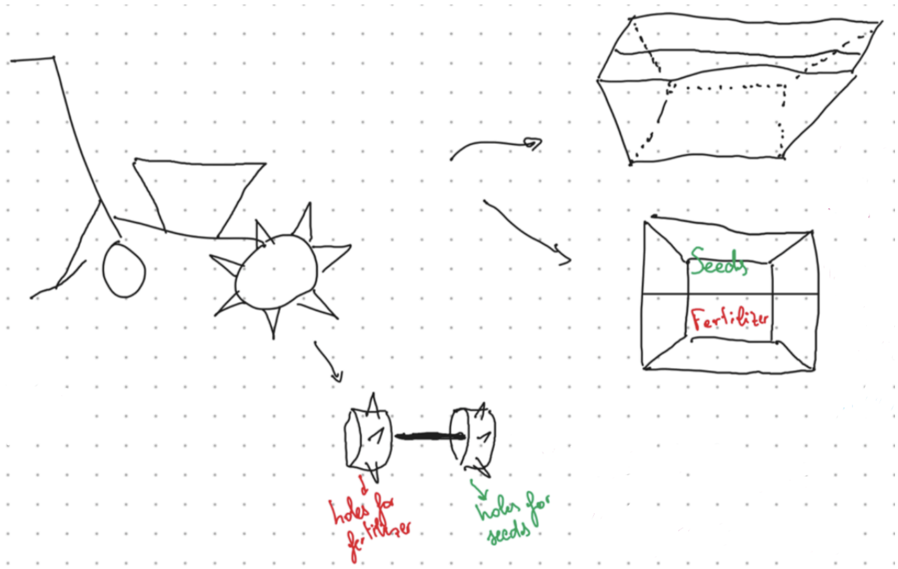
- Card 2: add motion = we added the rotating wheel so it can continue unlimitedly
- Card 29: create system = multi-stage process to complete the function
- Card 71: use human-generated power = using the kinetic of the user to push the machine without the help of a motor

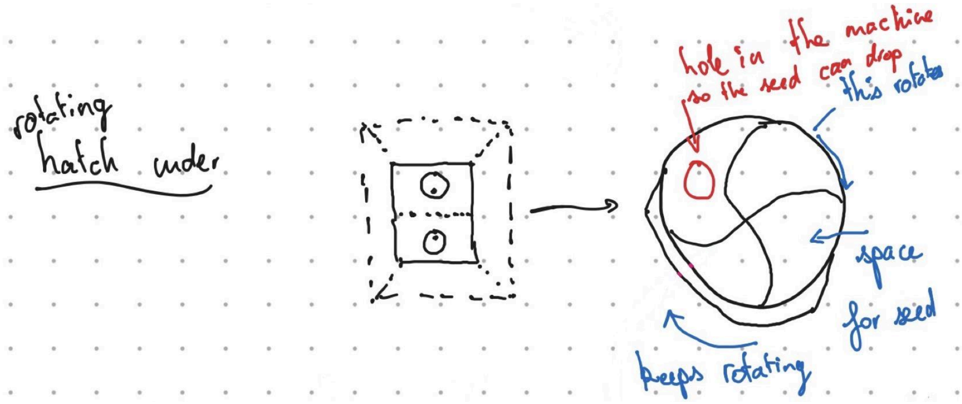
In the last update we used card 38 to impose hierarchy on functions:



1.4 Design D by Antoine

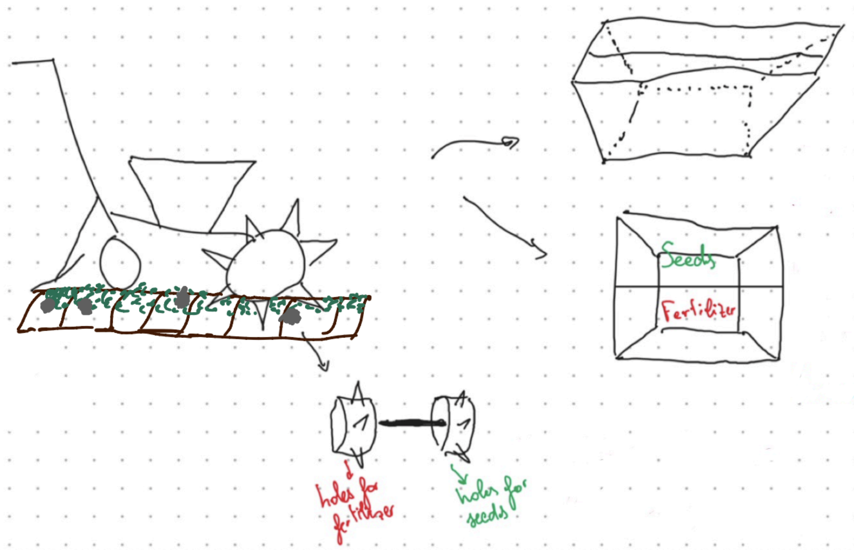
This concept is a wheelbarrow-like device equipped with two spiked wheels operating in sequence. The first wheel creates holes and deposits fertilizer, while the second wheel deposits seeds in a separate location. The system uses compartmentalized storage to keep seeds and fertilizer separated, and ensures their controlled placement in the soil as the device moves forward. The seed selection mechanism relies on this horizontally rotating wheel housed securely between a top hopper and a bottom floor plate. Seeds rest in the hopper and fall through a single upper entry hole into an empty pocket created by the S-shaped dividers on the rotating disc. As the flat wheel turns, it sweeps that single, isolated seed in a horizontal arc around the circular chamber. Once the pocket rotates to the completely opposite side, it aligns with a secondary exit hole in the bottom floor plate, allowing the accurately timed seed to drop directly down into the soil.





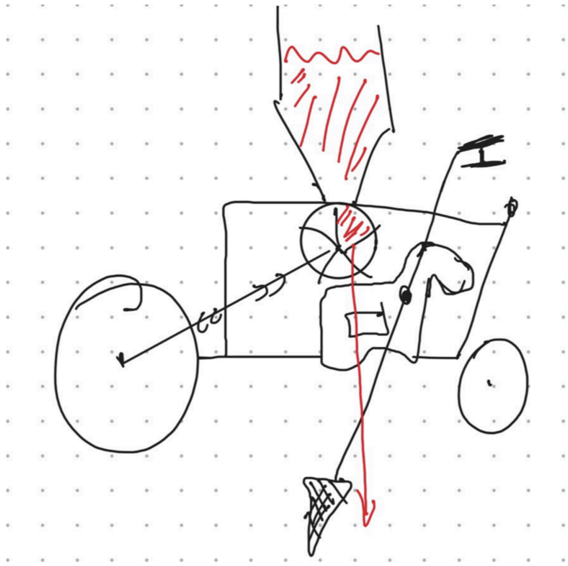
Heuristic Cards
- Card 23: compartmentalize = separate the seeds from the fertilizer
- Card 29: create system = multi-stage process to complete the function
- Card 38: impose hierarchy on functions = first digging the hole, then planting the seed and finally covering the hole
- Card 71: use human-generated power = using the kinetic of the user to push the machine without the help of a motor

In the last update we used card 39 to incorporate environment:



1.5 Design E by Alex

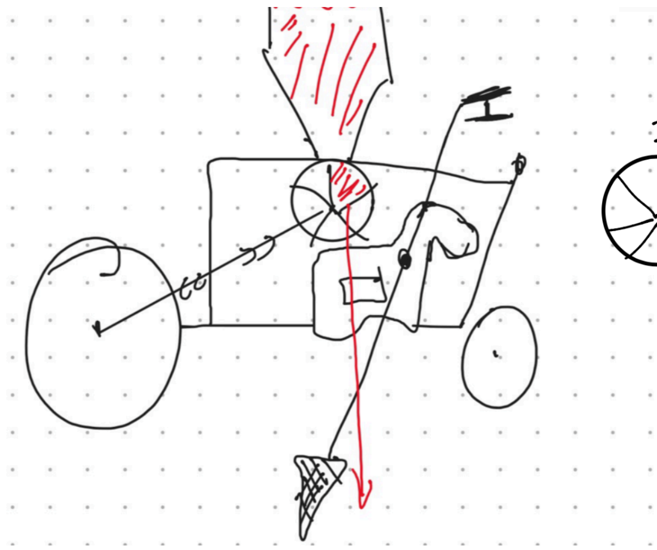
This concept is a cart-based system equipped with a manually actuated spear-like tool for soil penetration. The user drives the spear vertically into the ground to create a planting hole, then advances the cart to the next position. The spear is lifted, repositioned at an angle, and redeployed to repeat the process. Seeds and fertilizer are dispensed through an integrated dosing mechanism during each insertion cycle.

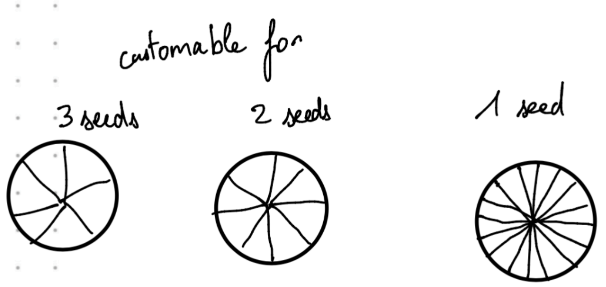


Heuristic Cards

- Card 14: attach independent functional components = we combined the 3 tools individually
- Card 29: create system = multi-stage process to complete the function
- Card 40: incorporate user input = user needs to push the handle so it can work
- Card 71: use human-generated power = using the kinetic of the user to push the machine without the help of a motor

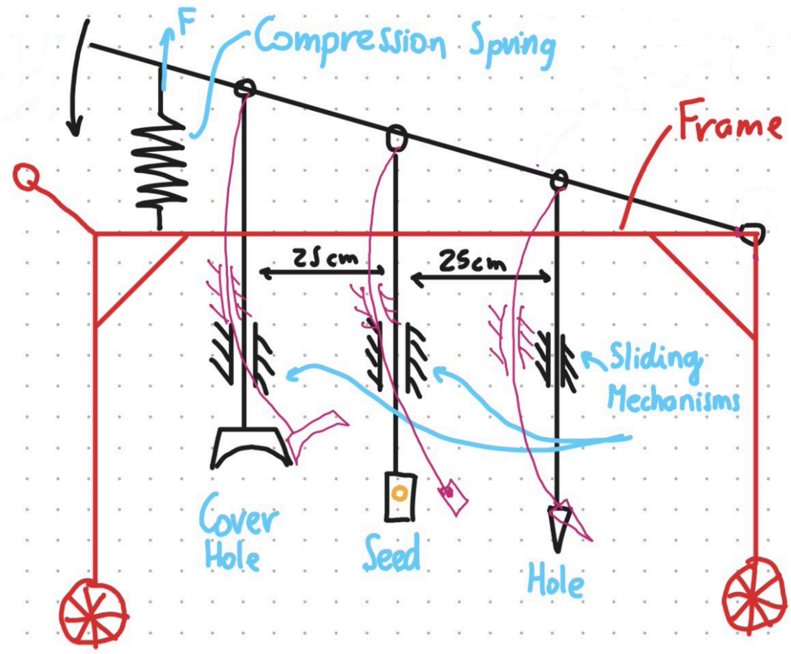
In the last update we used card 49 to offer optional components:





1.6 Design F by Luis

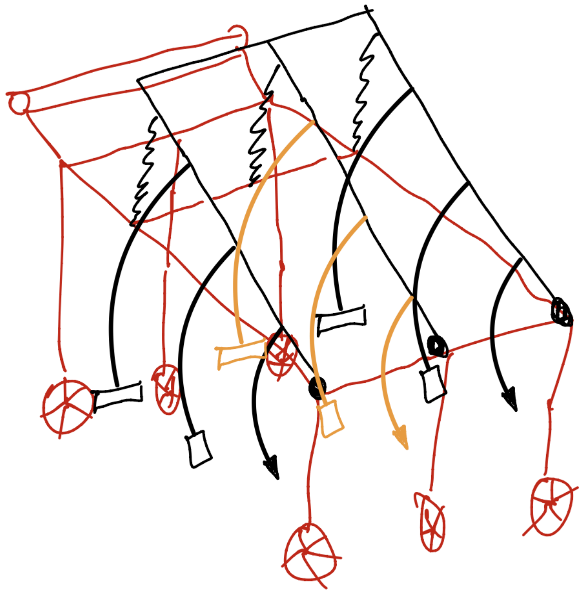
This concept is a four-wheeled cart integrating three separate vertically actuated mechanisms. The first mechanism creates the planting hole, the second deposits the seed, and the third covers the hole. Each function is sequentially activated by the user, enabling a clear multi-stage planting process within a single system.



Heuristic Cards

- Card 23: compartmentalize = separate distinctly the functions
- Card 29: create system = multi-stage process to complete the function
- Card 33: expose interior = inner parts are visible enabling the visual control of user
- Card 38: impose hierarchy on functions = first digging the hole, then planting the seed and finally covering the hole
- Card 40: incorporate user input = user needs to push the handle so it can work
- Card 71: use human-generated power = using the kinetic of the user to push the machine without the help of a motor

In the last update we used card 54 to repeat the function:



1.7 Reflect on the benefits and disadvantages

Brainwriting encourages the generation of a wide range of ideas in a short time and allows all team members to contribute equally. It promotes creativity by avoiding early judgment and helps explore different solution directions. However, since the process relies heavily on imagination, it can be difficult to fully integrate all user needs from the beginning. This often leads to simple or redundant concepts in the early stages.

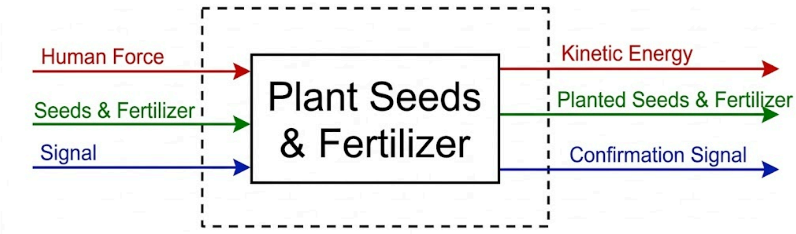
The use of heuristic cards significantly improves the process by structuring the design exploration. They help refine initial ideas, introduce new perspectives, and reveal hidden issues that were not identified during brainwriting. Additionally, they facilitate collaboration by incorporating different viewpoints within the team.

On the downside, the large number of heuristic cards makes it difficult to know when to stop the iteration process. Some combinations may also lead to overly complex solutions that are not practical. Overall, simple heuristic cards are more effective for generating new concepts, while more advanced ones are better suited for improving and refining existing designs.

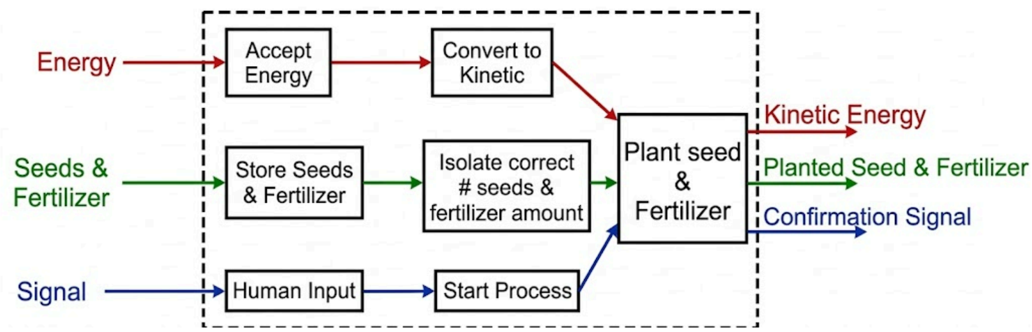
2. Concept Generation - Functional decomposition / Morphological Matrix

2.1 Functional Decomposition

Top Level Function



First level of decomposition



Second level of decomposition

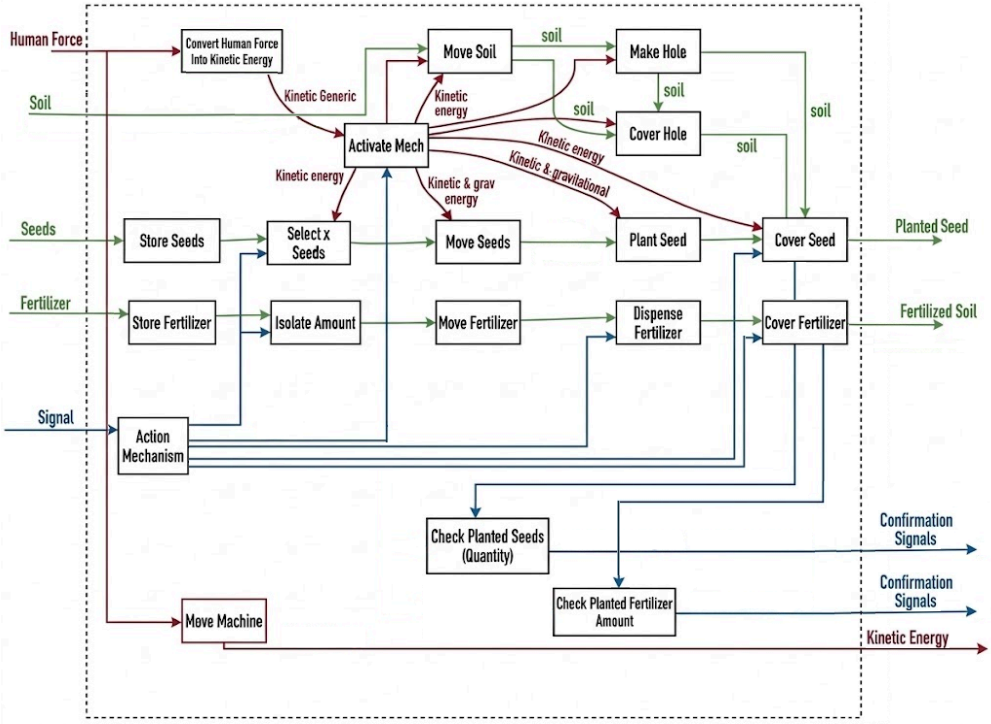


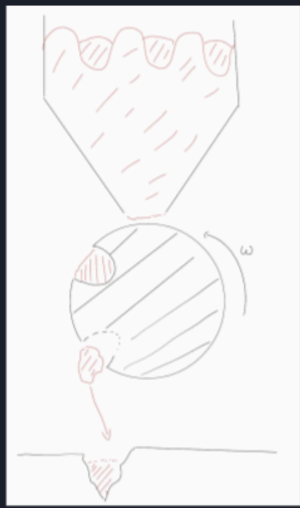
Table:

Store Seeds and Fertilizer	Isolate Correct amount	Deliver the Load
Hopper	Empty Drum with exact volume	Gravity
Bucket	Infinite Screw	Articulated Claw
Bags	Infinite Zig-Zag Saw	Push System

Concept 1:

Concept Combination Table

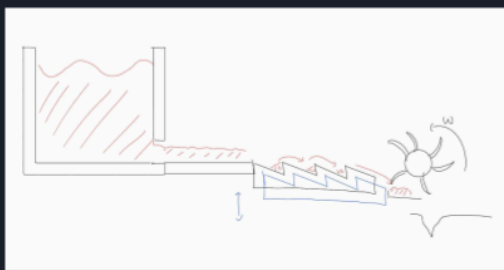
Store Seeds and Fertilizer	Isolate Correct amount	Deliver the Load
Hopper	Empty Drum with exact volume	Gravity
Bucket	Infinite Screw	Articulated Claw
Bags	Infinite Zig-Zag Saw	Push System



Concept 2:

Concept Combination Table

Store Seeds and Fertilizer	Isolate Correct amount	Deliver the Load
Hopper	Empty Drum with exact volume	Gravity
Bucket	Infinite Screw	Articulated Claw
Bags	Infinite Zig-Zag Saw	Push System



Concept 3:

Concept Combination Table

Store Seeds and Fertilizer	Isolate Correct amount	Deliver the Load
Hopper	Empty Drum with exact volume	Gravity
Bucket	Infinite Screw	Articulated Claw
Bags	Infinite Zig-Zag Saw	Push System

2.2 Reflection on Benefits and Disadvantages.

Benefits: This method forces the user to work in a more abstract and imaginative way to reach the intended function that was already determined in Hand-in 1. It is a great way to make bias disappear and make the user influence-free of ideas that might seem preferable. For this reason we can say it does have a positive impact on the process.
Disadvantages: However, using it correctly and as intended can be very time consuming and confusing. This is because not all of the products and design processes are the same, and applying a general framework to specific situations is not always the best solution. In the end, it can even lead to more confusions than improvements along the process.

In short, it can be undoubtedly the optimal tool for a big proportion of the design processes in the industry. However, one must not use it blindly and add its own critical thinking to see whether there might be a better option with similar results. Especially, in simpler machines where there is no specific transmission of signal/energy.
The table generator concept tool can also be very useful when many alternatives are available. In our case, due to the limited availability of energy sources and materials/techniques, we believe it was not as important to use as in other cases.

3. Concept Selection Process

3.1 Decision matrix method

The design concepts developed with the Brainwriting are evaluated. The ones from the Functional Decomposition will be used for detailed mechanisms or potential ideas .

To select the definitive concept the Pugh Concept Screening Decision Matrix is used. A reference form the market has been selected to compare the different concepts. [Link to the website for more information]

	CONCEPT VARIANTS						
Selection Criteria	**A**	**B**	**C**	**D**	**E**	**F**	**REF**
Ease of Operation	-	0	+	0	-	-	0
Number of Pieces	0	+	-	0	-	-	0
Weight	0	-	-	0	-	-	0
Manufacturing Ease	+	+	0	+	-	0	0
Portability	+	+	+	0	+	+	0
Ridge Conservation	+	0	+	0	+	+	0
Isolation of Seeds	-	0	0	-	-	+	0
Consistent spacing	-	-	0	0	-	-	0
Hole coverage	-	0	+	0	-	-	0
Hole Creation	0	-	0	0	0	+	0
Maintenance and Repair	+	0	0	-	-	+	0
Human Energy	-	0	0	-	-	-	0
Cost	+	-	0	-	-	-	0
Variable setups	+	+	0	0	0	+	0
PLUSES	6	4	4	1	2	6	
SAMES	3	6	8	9	2	1	
MINUSES	5	4	2	4	10	7	
	14	14	14	14	14	14	
NET	1	0	2	-3	-8	-1	
RANK	2	3	1	5	6	4	
CONTINUE?	YES	NO	YES	NO	NO	NO	

The two best ranked concepts are A & C.

A Pugh concept screening method was used to evaluate the six generated concepts based on the defined user needs. Concepts E and F were excluded early from the selection process due to their low feasibility: Concept E was considered inefficient, while Concept F was overly complex in terms of mechanism and operation.

Concept B presented an effective seed selection mechanism; however, it was discarded as a complete solution because the repeated opening and closing of the soil using blades was expected to require excessive force from the user. Nevertheless, its seed metering principle was identified as a valuable feature for further development.

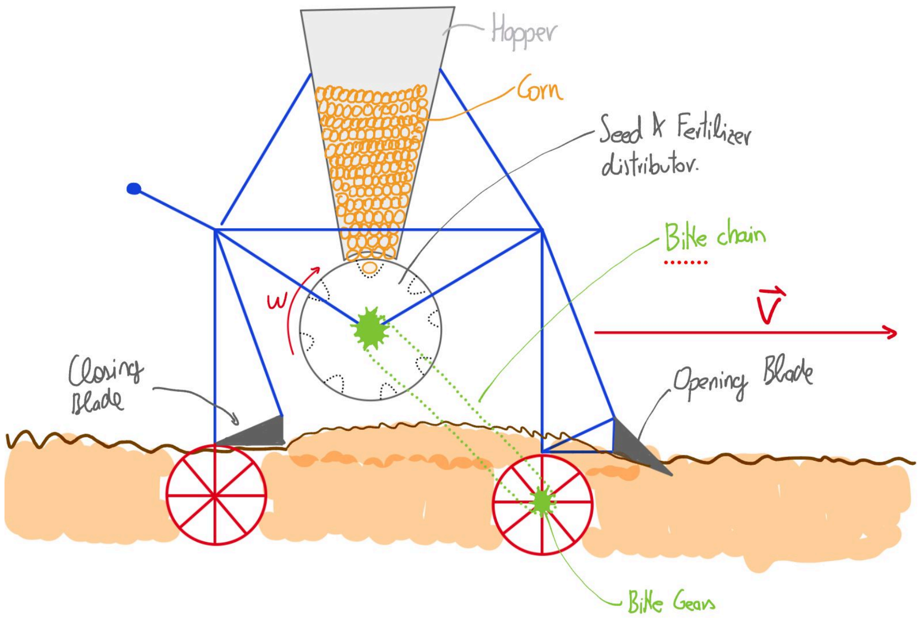
Concepts C and D both proposed cart-based systems, but Design D introduced parallel dispensing of seeds and fertilizer, which was considered unsuitable for ridge farming due to excessive width. A sequential arrangement was preferred. Concept C showed potential in its overall architecture, but its hole-opening mechanism was judged too complex and its seed selection system insufficiently defined.

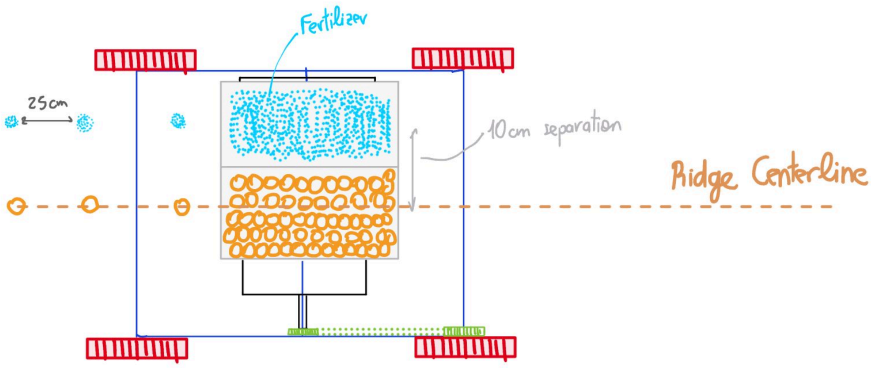
Concept A was retained for its robust and simple soil interaction using a front wheel to create holes. However, its manual seed and fertilizer actuation was seen as a limitation.

As a result, Concepts A and C were identified as the most promising. Concept A offers a simple and reliable soil interaction mechanism, while Concept C provides a strong cart-based structure with potential for continuous operation. These concepts were selected as a basis. Combining their strengths while addressing their respective weaknesses, after a long discussion the team created a very promising concept described below:

The hopper from concept A and the central wheel of concept C were combined. The drilling wheel of concept A was moved and was substituted by an opening blade from concept B. The same blade is used inversely to close the ridge. The hopper was designed to contain both the fertilizer and the corn seeds, as in concept D. To ensure a proper seeding distance, bike gears and a chain were used to connect the seed and fertilizer distributor with the wheels at a specific gear ratio. The distributor only allows one seed in each hole and 3-5 gr for the fertilizer. The seeds are then dropped every 25 cm.

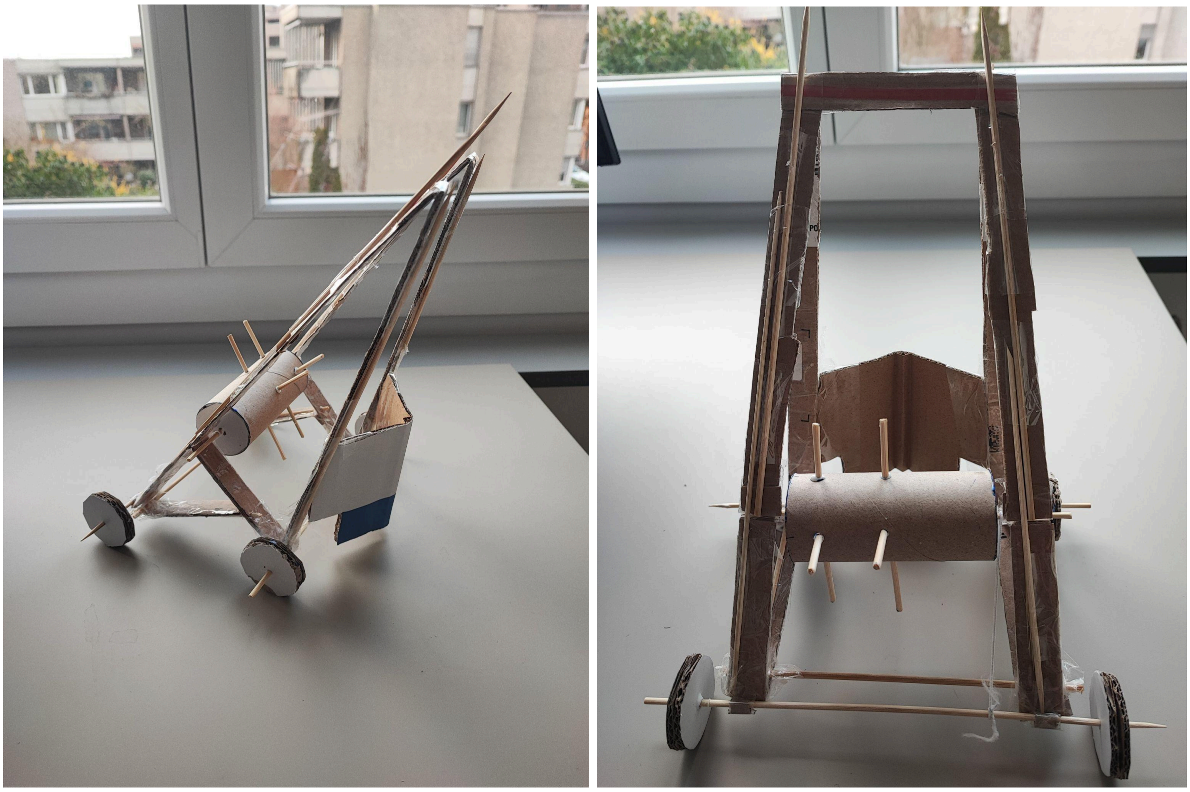
By combining the different concepts, a rather simple solution was obtained without complex mechanisms, and the key components, such as the bike chain and gears, can be found in Malawi.





3.2 Cardboard prototypes of the two most promising concepts

First prototype concept (C)



Figures: First cardboard prototype depicting the structure of concept C.

This cardboard prototype serves as an excellent physical representation of Design C, capturing the fundamental architecture of the precision seeder. It perfectly demonstrates the overall structure of the tool, establishing highly accurate proportions that give a clear sense of its physical footprint and handling mechanics. Most importantly, the model effectively visualizes the sequence of operations, specifically detailing how the rear blade is positioned. By placing this angled blade directly behind the main planting mechanism, the prototype clearly illustrates how the seeder will efficiently push displaced soil back over the track, filling the newly created holes and protecting the seeds as the tool moves forward.

However, building this model also highlights why the current seed-dropping system is too mechanically ambitious for the final product. The rotating cylinder relies on a complex series of spikes that must penetrate the soil perfectly. To work as intended, this system would require sophisticated internal gearing and highly precise timing mechanisms designed to push the spike into the dirt and simultaneously trigger a micro-hatch to open, dropping exactly one seed into the hole before closing again. This level of mechanical intricacy introduces too many potential points of failure and steepens the manufacturing difficulty significantly. Because of this, the final concept will pivot; it will retain the highly successful structural frame, wheel base, and rear blade from Design C, but will replace the spiked-cylinder mechanism with a more streamlined and reliable seed-dispensing solution.

Second prototype concept (C)

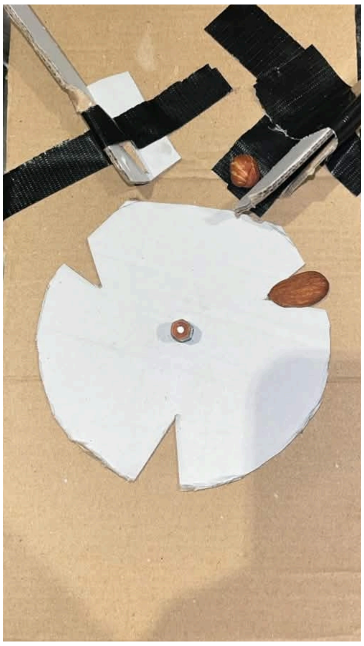


Figure: Second cardboard prototype for wheel mechanism

This cardboard prototype effectively demonstrates the new seed-spacing mechanism, designed to ensure perfectly even 25-centimeter intervals between plantings. Positioned securely between the main seed hopper and the ground, the system relies on a rotating dispensing wheel, represented by the white notched disc, to regulate the flow of seeds. As shown in the mockup, the seed container sits directly above this wheel, using gravity to feed the seeds downward. The edge of the wheel features specifically calibrated cutouts that act as single-seed pockets, sized perfectly to catch and isolate just one seed at a time (much like the almond shown resting perfectly in the right-side notch). As the planter moves forward, a bike chain drives the rotation of this wheel. This steady, mechanically linked rotation carries the isolated seed along the curve and drops it into the soil exactly every 25 centimeters, providing a highly reliable solution that will be part of the final concept.

3.3 Reflect on the pros and cons of the previous method

This method provides a clear evaluation framework, allowing concepts to be compared based on defined criteria rather than subjective scores. The use of a reference concept helps ensure consistent comparison, while the structured approach makes decision-making faster and reduces unnecessary debates within the team.

However, the results strongly depend on the chosen reference concept and the defined user needs. If these are incomplete or biased, the ranking may not be reliable. In addition, all criteria are treated equally, even though some are more important than others, which can lead to less accurate conclusions.

Overall, the method is efficient and useful for structured comparison, but it requires careful definition of inputs to ensure meaningful results.

4. Contributions

part 1 : All
part 2 : All
part 3 : All